Mohammad Al Hosan Kanon

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EDUCATION

B.Sc. (Hons.) **Bangladesh University of Engineering & Technology (BUET)** Dec 2021 – Jun 2026
Department of Electrical and Electronic Engineering
Major: Electronics and Photonics
CGPA: 3.78/4.00 (4.00/4.00, Senior year)

RESEARCH EXPERIENCE

• **Research Project** Feb 2026 – Aug 2026

Mitigation of Short-Channel Effects in 20 nm Double-Gate MOSFETs via Step Oxide, Dual-Material Gate, and Halo Doping Details

- *Supervisor: Dr. Sharif Mohammad Mominuzzaman, Professor and Head, Dept. of EEE, BUET*

  ◦ Achieved a 60% reduction in DIBL and nearly sixfold improvement in I_{ON}/I_{OFF} using step-oxide, dual-material-gate, and halo-doping engineering.

• **Undergraduate Thesis** Apr 2025 – Jun 2026

Schottky-to-Ohmic Contact Modulation in Graphene/Janus MoSeTe van der Waals Heterostructures via Electric Field and Biaxial Strain: A First-Principles Study Details

- *Supervisor: Dr. Sharif Mohammad Mominuzzaman, Professor and Head, Dept. of EEE, BUET*

  ◦ Identified multiple independent pathways to a fully Ohmic Graphene/MoSeTe contact via electric field and biaxial strain modulation of the Schottky barrier.

  ◦ Proposed a 2D field-effect transistor architecture exploiting the tunable contact behavior.

RESEARCH PUBLICATIONS

Journal | Thesis Extension

[1] **M. Al Hosan Kanon** and S. M. Mominuzzaman, "Schottky-to-Ohmic Contact Modulation in 1T-NbS_2/Janus 2H-MoSeTe van der Waals Heterostructures via Electric Field and Biaxial Strain: A First-Principles Study," In: *Nanoscale Advances* (Royal Society of Chemistry), *(Under Revision)*

Conference Proceedings

[1] **M. Al Hosan Kanon** and S. M. Mominuzzaman, "Mitigation of Short-Channel Effects in 20 nm Double-Gate MOSFETs via Step Oxide, Dual-Material Gate, and Halo Doping," In: *14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh, 2026. *(Under Review)*

[2] **M. Al Hosan Kanon**, S. A. Rifat, I. Ahmad, and M. A. Islam, "Design and Simulation of a Class-A Common-Emitter RF Power Amplifier for the 433 MHz ISM Band in Cadence Virtuoso," *Proc. 5th Int. Conf. Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, Bangladesh, 2026, pp. 1–6. Xplore

[3] **M. Al Hosan Kanon**, S. T. Islam, N. Sadik, S. A. Rifat, I. Ahmad, M. A. Islam, and A. Sarkar, "Synthesis Constraint Sweep and Physical Implementation of a Parameterized MAC Unit: Cross-Toolchain Validation on Commercial 45 nm and Open-Source 130 nm Processes," In: *14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh, 2026. *(Under Review)*

[4] **M. Al Hosan Kanon**, M. M. Rahman, P. Mobarak, and M. Al Amin, "IoT-Based Real-Time Monitoring and Preservation System for Small-Scale Fruit Storage," *Proc. 5th Int. Conf. Electrical, Computer & Telecommunication Engineering (ICECTE)*, Rajshahi, Bangladesh, 2026, pp. 1–6. Xplore

[5] **M. Al Hosan Kanon**, I. Ahmad, D. A. Saha, M. A. Amin, N. Tasnim, and P. Mobarak, "Design and Implementation of a Dual-Mode Floor Mopping Robot," *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh, 2025, pp. 421–426. Xplore

[6] M. Al Amin, **M. Al Hosan Kanon**, and M. M. Rahman, "Bridging Spectral Specificity and Semantic Generalization for Respiratory Sound Classification: A Comparative Study of MFCCs and Augmented YAMNet Embeddings," *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1–6. Poster | Xplore

SELECTED PROJECTS

RTL-to-GDSII Implementation of an Optimised Multiply-Accumulate Unit 2026
Tools: SystemVerilog, Cadence Genus/Innovus (GPDK045), OpenROAD (IHP SG13G2 130nm)

Optical Quality Analysis and Classification of LEDs Using KNN and Random Forest Models 2026
Tools: Python (scikit-learn, pandas), Streamlit, Arduino (BPW34 Photodiode, TCS34725 Sensor, PySerial)

Smart Grid Type UPS Using Bidirectional Inverter 2024–2025
Tools: Push-Pull Inverter Topologies, Solar Charger, DC Battery Storage

UART Design Using Verilog for FPGA Implementation 2024–2025
Tools: Verilog HDL, Moore FSM, CPLD/FPGA

WORK EXPERIENCE

Lecturer in Physics — **Academic & Admission Pathshala** [Website] — Nov 2022 – Nov 2024
Part-time, Online — *Dhaka, Bangladesh*
- Taught university-admission and secondary physics.
- Developed comprehensive lectures.

Intern — **Inovace Technologies Ltd.** [Website] [Credential] — Nov 2025
In-Person — *Dhaka, Bangladesh*
- Collaborated with R&D on biometric devices.
- Worked on hardware–software integration.
- Analyzed 'Tipsoi' product architecture.
- Observed industrial QC and RMA workflows.

LANGUAGE PROFICIENCY

IELTS Academic: Overall **7.0** (CEFR C1), Listening: 7.5 | Reading: 8.0 | Writing: 6.5 | Speaking: 6.5

TECHNICAL EXPERTISE

Materials & Device Simulation: Silvaco ATLAS (2D; DeckBuild, TonyPlot), Quantum ESPRESSO (PWscf, PHonon, PostProc), Materials Studio (DMol3), VESTA.
EDA & Circuit Design: Cadence (Virtuoso, Spectre; Genus, Innovus), Intel Quartus, LTspice, Proteus.
Hardware: FPGA, Arduino, STM32.
Programming Languages: Python, MATLAB, C/C++, Verilog/SystemVerilog (HDL), HTML.
Documentation & Design: LaTeX (Overleaf, TeXstudio), MS Word, MS PowerPoint, Adobe Illustrator.

AWARDS & SCHOLARSHIPS

[1] Dean's List Award 2026
Bangladesh University of Engineering & Technology (BUET)

[2] University Merit Scholarship 2025 – 2026
Bangladesh University of Engineering & Technology (BUET)

[3] University Stipend Scholarship 2025
Bangladesh University of Engineering & Technology (BUET)

[4] Government Board Scholarship (General), SSC 2018
Board of Intermediate and Secondary Education, Chattogram

NON-CREDIT FABRICATION COURSE

Semiconductor Fabrication 101 2026
Purdue University, The University of Texas at Austin & Intel
- Covered lithography, oxidation, diffusion, ion implantation, deposition, etching, and packaging.
- Trained on vFabLab, a virtual cleanroom simulator.

PROFESSIONAL SERVICE & MEMBERSHIPS

Peer Reviewer (Journal) | *Physica Scripta* (IOP Publishing) Present
Peer Reviewer (Conference) | *IEEE COMPAS* (IEEE CS BDC, University of Dhaka) 2026
Memberships | *IEEE* (Student Member), *IEEE Young Professionals*, *BUET Robotics Society*